

Harmonic Gaps, Gamma Derivatives, and Mixed Tail Moments

Joint Lerch generators and exact centered values

11 October 2026

Repository baseline. 7bd45777a8a127e5dc69aff8887ca36a79a3059d. The manuscript and all eleven archives in the incoming directory were inspected at this revision.

Abstract

We address three connected questions in the newest harmonic-parity report. First, arbitrary zero blocks around any finite number of independently moving harmonic orders admit one jointly entire multiple-Lerch completion. A confluent divided-difference kernel gives a common tube of holomorphy; the complete decoration series converges on the unit polydisc, independently of the number of free orders. Second, a derivative transverse to a nonpositive harmonic index is represented by an explicit convergent logarithmic Mellin integral and by a finite difference of decorated lower-depth sums. At a zero index the first new gap is a log-Gamma difference; higher derivatives give normalized Stieltjes primitives. All derivative orders, all nonpositive centers, and arbitrary unmarked nonpositive decorations are included. Third, every centered even moment of a product of two even-order Hurwitz tails is evaluated by a finite Bernoulli formula. Classical odd-weight Euler reduction puts the values in an explicit finite span of ordinary zeta products. One analytic generating function produces the entire moment family and has finite polylogarithmic primitives.

The underlying harmonic interpolant, its untwisted geometric kernel, and the classical Euler and Hurwitz identities are credited to their sources. The contribution is the stated extension of the repository's exact questions, not a claim of literature-wide priority. Proofs are separated from exact finite checks and high-precision diagnostics. The Gaussian S_6 and revised S_8 evaluations remain conjectural.

Contents of the accompanying package. Modular and standalone L^AT_EX sources, this PDF, exact verification programs, numerical diagnostics, result records, source provenance, an integration ledger, and further research questions.

Contents

1	Scope, principal results, and conventions	1
1.1	What is established	1
1.2	Provenance and the precise claim of advancement	2
1.3	Analytic and algebraic conventions	2
2	A jointly entire kernel for all zero-block decorations	3
2.1	The established geometric kernel in divided-difference form	3
2.2	Why the kernel gives the specified noninteger interpolant	4
2.3	The entire multiple-Lerch completion	5
2.4	Summing every zero block on a dimension-independent polydisc	6
3	Transverse index derivatives and Gamma–Stieltjes gaps	7
3.1	The joint Hurwitz completion	8
3.2	A finite gap identity before specialization	9
3.3	Gamma gaps, Stieltjes primitives, and repeated primitives	11
3.4	Absolutely convergent transverse Mellin formulas	12
3.5	All negative outer values and fixed Barnes constants	13
3.6	A finite family after arbitrary unmarked deletions	14
4	Mixed centered harmonic-tail moments	15
4.1	Continuation and ordinary convergent representatives	15
4.2	A finite formula at every moment order	17
4.3	Reduction to ordinary zeta values	19
4.4	The first unequal-order family	20
4.5	An analytic generator for all moment orders	20
4.6	Explicit polylogarithmic primitives and their constants	21
4.7	Scope of the result	22
5	Source audit, normalization safeguards, and verification	23
5.1	Inspected baseline and source attribution	23
5.2	An absolutely convergent boundary series can have a different value	23
5.3	Hypotheses and constants to preserve during integration	24
5.4	Reproducible verification and the limits of its evidence	25
5.5	Status of the broader conjectures	26
6	Further exact research questions	26
6.1	Higher products and logarithmic moments	26
6.2	Several moving positions and primitive calculus	27
6.3	Rational shifts and continuation of the generators	28
6.4	Integration with the existing conjectural program	29

1 Scope, principal results, and conventions

This continuation starts at the explicit questions R1, R7, and R8 of *Harmonic Parity and Resolvent Identities*, preserved in the incoming directory at the pinned revision [2, 1]. R1 asks for mixed even-tail moments, beginning with orders two and four. R7 asks for a genuinely convergent generator with several free harmonic orders and arbitrary zero blocks. R8 asks for a derivative transverse to a harmonic index that is subsequently set to a nonpositive integer. The corresponding results here are independent theorems with compatible normalizations.

1.1 What is established

- 1. All zero decorations with independent orders.** Theorem 2.4 proves a single analytic formula for

$$\sum_{p_0, \dots, p_k \geq 0} \mathcal{H}_{a,b}(0^{p_0}, s_1, 0^{p_1}, \dots, s_k, 0^{p_k}) u_0^{p_0} \cdots u_k^{p_k}.$$

Every s_j is an independent complex variable. The series converges locally normally for $|u_j| < 1$, with a domain that does not shrink with k . The proof gives all branch cancellations and does not infer an analytic identity from agreement at integer endpoints.

- 2. Transverse Gamma and Stieltjes gaps.** Theorems 3.2 and 3.5 prove every order of

$$\partial_u^r \mathcal{H}_{a,b}(s, u, t)|_{u=-m}, \quad m \geq 0, \quad r \geq 1.$$

The first new case is the log-Gamma gap $\log \Gamma(z+1) - \log \Gamma(y)$ between two surviving lattice variables $y > z$. There are explicit absolutely convergent Mellin integrals, finite decorated-sum formulas, and a finite family sufficient after arbitrary unmarked nonpositive indices have been removed. The simultaneous Stieltjes completion at $u = 1$ is included.

- 3. Every mixed even-tail moment.** Section 4 evaluates all centered even moments of $(\zeta(A) - H_n^{(A)})(\zeta(B) - H_n^{(B)})$ for even $A, B \geq 2$. The finite formula uses Bernoulli numbers and classical double zeta values of odd weight; Euler's reduction removes the latter. For each fixed pair (A, B) a fixed finite span contains the entire infinite sequence of moments. An analytic generating function and finite ordinary-polylogarithm primitives complete the result. This resolves the two-factor sector of R1; products with additional harmonic factors are not covered by the same theorem.

For a concrete illustration, put $x_n = n + \frac{1}{2}$ and $T_j(n) = \zeta(j, n+1)$. The unequal-order identities include

$$\sum_{n \geq 0} T_2(n) T_4(n) = \frac{15}{2} \zeta(5) - 3 \zeta(2) \zeta(3), \quad (1.1)$$

$$\sum_{n \geq 0} x_n^2 T_2(n) T_4(n) = \frac{1}{4} \zeta(2) \zeta(3) + \frac{1}{6} \zeta(2) + \frac{1}{2} \zeta(3) - \frac{5}{8} \zeta(5), \quad (1.2)$$

$$\sum_{n \geq 0} \left\{ x_n^4 T_2(n) T_4(n) - \frac{1}{3} \right\} = -\frac{7}{80} \zeta(2) \zeta(3) - \frac{1}{30} \zeta(2) - \frac{1}{4} \zeta(3) + \frac{7}{32} \zeta(5) - \frac{3}{40}. \quad (1.3)$$

All three displayed series, including the brackets in the last one, are ordinary convergent series. The third identity must retain its subtraction. Since $T_2(n) = \psi'(n+1)$ and $T_4(n) = \psi'''(n+1)/6$, multiplication by six gives corresponding polygamma-product evaluations.

1.2 Provenance and the precise claim of advancement

The harmonic interpolant and its original geometric Mellin kernel are established mathematics. Ihara, Nakamura, and Yamamoto prove their entire interpolation and harmonic product law; their paper also explains related earlier work of Komori [5, Proposition 3.3, Theorem 4.1, Corollary 4.2]. The incoming independent-orders report identifies the same interpolant and gives a relative endpoint normalization [3]. The zero-index polynomial reductions and one-free-order generator are already in [2]. We retain those definitions and give the requested simultaneous analytic extension and transverse formulas.

Similarly, Bernoulli power sums, Hurwitz specializations, Gamma asymptotics, and Euler's odd-weight double-zeta reduction are classical [7, 8, 9, 6]. They are ingredients of the mixed-tail theorem, rather than separate claims of new mathematics.

The present results are proposed additions relative to the inspected repository. We make no assertion of literature-wide priority for every specialization, no arithmetic-independence claim for a spanning set, and no proof-assistant formalization claim. The original S_6 and revised S_8 candidates are not proved or refuted here.

1.3 Analytic and algebraic conventions

All endpoint parameters a, b lie in the right half-plane. Principal logarithms define $(n+a)^{-s}$. Positive real integration variables use their real logarithms. The Hurwitz zeta function is normalized by

$$\zeta(1+v, x) = \frac{1}{v} + \sum_{r=0}^{\infty} \frac{(-1)^r}{r!} \gamma_r(x) v^r, \quad \gamma_0(x) = -\psi(x). \quad (1.4)$$

A superscript $\zeta^{(r)}$ denotes differentiation in the first argument. In particular [7, §25.11],

$$\zeta(-m, x) = -\frac{B_{m+1}(x)}{m+1}, \quad \zeta'(0, x) = \log \Gamma(x) - \frac{1}{2} \log(2\pi). \quad (1.5)$$

Bernoulli numbers mean $B_j = B_j(0)$, so $B_1 = -1/2$ and $B_1(1) = 1/2$. The strict multiple-zeta convention is

$$\zeta(r, t) = \sum_{n > j \geq 1} n^{-r} j^{-t}.$$

For a word $w = (s_1, \dots, s_k)$, define the meromorphic ordered tail

$$Z_c(w) = \sum_{n_1 > \dots > n_k \geq 0} \prod_j (n_j + c)^{-s_j}, \quad Z_c(\emptyset) = 1.$$

An initial domain with $\Re s_j > 1$ suffices below. The relative interpolant is the unique triangular solution of

$$Z_b(w) = \sum_{w=uv} Z_a(u) \mathcal{H}_{a,b}(v), \quad \mathcal{H}_{a,b}(\emptyset) = 1. \quad (1.6)$$

The sum includes both empty prefix and empty suffix. Its depth-one value is

$$h_{a,b}(s) = \zeta(s, b) - \zeta(s, a). \quad (1.7)$$

The pole at $s = 1$ cancels. Useful values are

$$\begin{aligned} h_{a,b}(1) &= \psi(a) - \psi(b), & h_{a,b}^{(r)}(1) &= (-1)^r \{\gamma_r(b) - \gamma_r(a)\}, \\ h'_{a,b}(0) &= \log \Gamma(b) - \log \Gamma(a), & h_{a,b}(0) &= a - b. \end{aligned} \quad (1.8)$$

At $a = b + N$, $N \in \mathbb{N}_0$, the relative word is the finite strict sum over $N > n_1 > \cdots > n_k \geq 0$. For general endpoints it means (1.6), with the joint continuation proved again in Section 2. The finite endpoint property alone is not our definition of continuation.

We use a cutoff constant only for a function with a specified asymptotic expansion in powers and logarithms of that cutoff. It is the coefficient of the monomial $N^0(\log N)^0$. Section 4 proves when that prescription agrees with the spectral value. This distinction is essential: even an accidentally convergent paired boundary series need not equal the analytic continuation from its initial half-plane.

2 A jointly entire kernel for all zero-block decorations

The one-free-order zero-block generator in the incoming harmonic-parity report admits a uniform extension to any number of independently moving orders. The extension is not obtained by interpolating finite sums from integer endpoint differences. A divided-difference identity proves the required analytic relation at arbitrary endpoints before any coefficient is extracted.

The underlying geometric kernel and its normalized Mellin interpolation are already present in Ihara, Nakamura, and Yamamoto [5, Proposition 3.3 and Theorem 4.1]; the published paper is *The Ramanujan Journal* **67** (2025), article 1, doi:10.1007/s11139-025-01045-2. In their notation the kernel below is $e^{-b \sum_j x_j} G_k(e^{-x_1}, \dots, e^{-x_k}; a - b)$. Equivalently it is $e^{-(b-1) \sum_j x_j}$ times their $b = 1$ kernel. Its divided-difference form, composition proof, and continuation proof are included to make the normalization and later estimates explicit; the underlying interpolant and its geometric Mellin representation are not claimed as new. The developments here are the simultaneous twist tube, the full dimension-independent zero-decoration polydisc, the all-block identity with arbitrary independent free orders, and its compatibility with the transverse derivative calculus developed below.

Write $\mathcal{H}_{a,b}(s_1, \dots, s_k)$ for the entire relative Hurwitz interpolant, normalized by

$$Z_b(w) = \sum_{w=uv} Z_a(u) \mathcal{H}_{a,b}(v), \quad Z_c(s_1, \dots, s_j) = \sum_{n_1 > \cdots > n_j \geq 0} \prod_{\ell=1}^j (n_\ell + c)^{-s_\ell}.$$

Empty words have value one. Throughout this section $\Re a, \Re b > 0$, and powers of the positive-real-axis variables use their real logarithms. All endpoint powers use the branches obtained on the right half-plane.

2.1 The established geometric kernel in divided-difference form

For $\mathbf{x} = (x_1, \dots, x_k)$ put

$$X_0 = 0, \quad X_j = x_1 + \cdots + x_j, \quad Q_j = e^{-X_j}, \quad \Delta = a - b.$$

At pairwise distinct nodes define

$$\mathcal{K}_{a,b}^{(k)}(\mathbf{x}) = e^{-bX_k - \sum_{j=1}^{k-1} X_j} \sum_{j=0}^k \frac{e^{-\Delta X_j}}{\prod_{\ell \neq j} (Q_j - Q_\ell)}. \quad (2.1)$$

The sum is the divided difference $[Q_0, \dots, Q_k]z^\Delta$, where the value at Q_j is specified as $e^{-\Delta X_j}$. This specification keeps track of the logarithms even when the exponential nodes wind around zero. Set $\mathcal{K}^{(0)} = 1$.

Lemma 2.1 (Kernel analyticity and exponential control). *The expression (2.1) extends holomorphically through every hyperplane $X_j - X_i = 0$, away from the nonzero-period hyperplanes below. Its only possible finite singular hyperplanes are*

$$X_j - X_i = 2\pi i\ell, \quad 0 \leq i < j \leq k, \quad \ell \in \mathbb{Z} \setminus \{0\}.$$

Define the open convex tube

$$\Omega_k = \left\{ \mathbf{t} \in \mathbb{C}^k : \left| \Im \sum_{j=p}^q t_j \right| < 2\pi \quad (1 \leq p \leq q \leq k) \right\}. \quad (2.2)$$

On compact subsets of $\Re a, \Re b > 0$ and $\mathbf{t} \in \Omega_k$, for every multi-index $\boldsymbol{\nu}$ there are $C, \delta > 0$ such that

$$\left| \partial_{\mathbf{x}}^{\boldsymbol{\nu}} \mathcal{K}_{a,b}^{(k)}(\mathbf{x} - \mathbf{t}) \right| \leq C e^{-\delta(x_1 + \dots + x_k)}, \quad x_j \geq 0. \quad (2.3)$$

Proof. If nodes collide with $X_j - X_i = 0$, their assigned logarithms agree. On a sufficiently small neighborhood of each cluster, choose the common holomorphic logarithm. The ordinary confluent divided difference of its holomorphic power function extends across all coinciding nodes. Different clusters have nonzero denominator differences. This proves the simultaneous removability assertion, including intersections of arbitrarily many zero hyperplanes away from the remaining singular hyperplanes. Nonzero exponential periods are exactly the remaining possible denominator zeros.

The first-order kernel is

$$\mathcal{K}_{a,b}^{(1)}(v) = \frac{e^{-bv} - e^{-av}}{1 - e^{-v}}.$$

The elementary divided-difference recurrence gives, for $k \geq 2$,

$$\mathcal{K}_{a,b}^{(k)}(v, w, \mathbf{z}) = \frac{e^{-v} \mathcal{K}_{a,b}^{(k-1)}(v + w, \mathbf{z}) - e^{-av} \mathcal{K}_{a,b}^{(k-1)}(w, \mathbf{z})}{1 - e^{-v}}. \quad (2.4)$$

Indeed the two divided differences in the numerator omit Q_1 and Q_0 , respectively; scaling the nodes of the second one by Q_1 gives exactly the displayed exponential factor.

This recurrence proves (2.3) by induction. For large x_1 , the denominator is uniformly bounded away from zero, and both numerator terms have the required exponential decay. For bounded x_1 , the only possible denominator zero is the removable zero at $v = 0$; the numerator vanishes there because its two kernel terms agree. A fixed small neighborhood of the shifted positive orthant stays away from all nonzero-period hyperplanes, because the compact subset in (2.2) has a positive distance from its boundary. The maximum principle, or the Cauchy formula around the removable zero, then bounds the quotient by the inductively bounded numerator. Merging $v + w$ preserves the tube inequalities, since they involve consecutive sums. One may take δ smaller than the minimum of $1, \Re a, \Re b$ on the endpoint compact set. Cauchy estimates in the same uniform neighborhood prove the bounds for every derivative. $\square$

2.2 Why the kernel gives the specified noninteger interpolant

Let

$$\mathcal{B}_c^{(k)}(\mathbf{x}) = \frac{e^{-cX_k - \sum_{j=1}^{k-1} X_j}}{\prod_{j=1}^k (1 - e^{-X_j})}, \quad x_j > 0. \quad (2.5)$$

This is the geometric kernel of the ordered infinite sum Z_c : introducing the positive successive gaps gives $\mathcal{B}_c^{(k)} = \sum_{n_1 > \dots > n_k \geq 0} e^{-\sum_j (n_j + c)x_j}$.

Lemma 2.2 (Exact endpoint composition). *For arbitrary right-half-plane endpoints a, b, c ,*

$$\mathcal{K}_{a,b}^{(k)}(x_1, \dots, x_k) = \sum_{j=0}^k \mathcal{K}_{a,c}^{(j)}(x_1, \dots, x_j) \mathcal{K}_{c,b}^{(k-j)}(x_{j+1}, \dots, x_k). \quad (2.6)$$

Consequently,

$$\mathcal{B}_b^{(k)} = \sum_{j=0}^k \mathcal{B}_a^{(j)} \mathcal{K}_{a,b}^{(k-j)}, \quad (2.7)$$

with the same prefix/suffix arguments.

Proof. Apply the divided-difference product identity

$$[Q_0, \dots, Q_k](fg) = \sum_{j=0}^k [Q_0, \dots, Q_j]f [Q_j, \dots, Q_k]g$$

to $f(z) = z^{a-c}$ and $g(z) = z^{c-b}$. The suffix nodes in its kernel are Q_ℓ/Q_j . Scaling a divided difference of order $k-j$ supplies the factor $Q_j^{k-j-(c-b)}$; together with the kernel prefactors this makes every summand have the common factor $e^{-bX_k - \sum_{\ell=1}^{k-1} X_\ell}$. Thus the product identity is exactly (2.6). This argument is first made at distinct positive nodes and then continued through the removable collisions.

For $x_j > 0$, let a real upper endpoint A tend to $+\infty$ in $\mathcal{K}_{A,b} = \mathcal{K}_{A,a} * \mathcal{K}_{a,b}$. In (2.1), every term except $j = 0$ tends to zero. The remaining term is precisely (2.5). This gives (2.7), a finite identity requiring no interchange with an infinite depth sum. In particular, the proof is valid for noninteger $a - b$ and does not use uniqueness of interpolation from integer values. $\square$

2.3 The entire multiple-Lerch completion

For $\Re s_j > 0$ define

$$\mathcal{E}_{a,b}(\mathbf{s}; \mathbf{t}) = \frac{1}{\prod_{j=1}^k \Gamma(s_j)} \int_{(0,\infty)^k} \prod_{j=1}^k x_j^{s_j-1} \mathcal{K}_{a,b}^{(k)}(\mathbf{x} - \mathbf{t}) d\mathbf{x}, \quad \mathbf{t} \in \Omega_k. \quad (2.8)$$

Theorem 2.3 (Independent spectral orders and simultaneous branch cancellation). *The function (2.8) extends jointly to an entire function of $\mathbf{s} \in \mathbb{C}^k$, holomorphic in $\mathbf{t} \in \Omega_k$, with locally normal derivatives of every fixed spectral and decoration order. It satisfies*

$$\mathcal{E}_{a,b}(\mathbf{s}; \mathbf{0}) = \mathcal{H}_{a,b}(\mathbf{s}), \quad \partial_{t_j} \mathcal{E}_{a,b}(\mathbf{s}; \mathbf{t}) = \mathcal{E}_{a,b}(\mathbf{s} - \mathbf{e}_j; \mathbf{t}). \quad (2.9)$$

In a neighborhood of $\mathbf{t} = \mathbf{0}$ its normally convergent Taylor series is

$$\mathcal{E}_{a,b}(\mathbf{s}; \mathbf{t}) = \sum_{\nu \geq 0} \mathcal{H}_{a,b}(\mathbf{s} - \nu) \frac{\mathbf{t}^\nu}{\nu!}. \quad (2.10)$$

For instance, the Taylor series converges normally whenever $\sum_j |t_j| < 2\pi$.

Proof. Lemma 2.1 proves absolute convergence of the integral. Repeated integration by parts gives the particularly useful continuation formula

$$\mathcal{E}_{a,b}(\mathbf{s}; \mathbf{t}) = \frac{(-1)^{|\mathbf{N}|}}{\prod_j \Gamma(s_j + N_j)} \int_{(0,\infty)^k} \prod_j x_j^{s_j+N_j-1} \partial_{\mathbf{x}}^{\mathbf{N}} \mathcal{K}_{a,b}^{(k)}(\mathbf{x} - \mathbf{t}) d\mathbf{x}, \quad (2.11)$$

valid for $\Re s_j > -N_j$. It agrees with the original integral where $\Re s_j > 0$; the endpoint terms there vanish. Since each N_j is arbitrary, these compatible formulas prove joint entireness. The same exponential majorants, with logarithmic powers from spectral differentiation, prove local normality.

At $\mathbf{t} = 0$, apply the normalized Mellin transform to (2.7), initially with all spectral real parts sufficiently large. The product of prefix/suffix transforms is a product of independent integrals. The result is exactly the defining deconcatenation relation for $\mathcal{H}_{a,b}$. This proves the first identity in (2.9); entireness extends it everywhere. The equality $\partial_{t_j} \mathcal{K} = -\partial_{x_j} \mathcal{K}$ and one integration by parts prove the lowering relation first where $\Re s_j > 1$, and then everywhere. Taylor's theorem proves (2.10). Every polydisc whose coordinate radii have sum less than 2π lies in Ω_k , giving the stated domain. $\square$

For $\mathbf{t} \in \Omega_k$ with $\Re t_j < 0$ for every j , let

$$\mathcal{Z}_c(\mathbf{s}; \mathbf{t}) = \sum_{n_1 > \dots > n_k \geq 0} \prod_j (n_j + c)^{-s_j} e^{(n_j + c)t_j}.$$

These ordered multiple-Lerch series (with the standard depth-one normalization [11, §25.14]) are absolutely convergent and entire in their independent orders. In that domain, $\mathcal{E}_{a,b} = \mathcal{Z}_a^{-1} * \mathcal{Z}_b$ under deconcatenation, by the same kernel identity. At each fixed depth the inverse is a finite expression. For example,

$$\begin{aligned} \mathcal{E}_{a,b}(s, t; u, v) &= \mathcal{Z}_b(s, t; u, v) - \mathcal{Z}_a(s, t; u, v) \\ &\quad - \mathcal{Z}_a(s; u)(\mathcal{Z}_b(t; v) - \mathcal{Z}_a(t; v)). \end{aligned}$$

Theorem 2.3 proves that the whole expression has a single holomorphic continuation throughout Ω_k , including all simultaneous zero-twist branch crossings. Continuing the summands independently without this completion is unnecessary.

2.4 Summing every zero block on a dimension-independent polydisc

Let $\mathbf{u} = (u_0, \dots, u_k)$, $|u_j| < 1$, and take the logarithms $L_j = \log(1 + u_j)$ determined at the origin. Put $t_j = L_j - L_{j-1}$.

Theorem 2.4 (All zero decorations around independent orders). *For every $k \geq 1$,*

$$\begin{aligned} \sum_{p_0, \dots, p_k \geq 0} \mathcal{H}_{a,b}(\{0\}^{p_0}, s_1, \{0\}^{p_1}, \dots, s_k, \{0\}^{p_k}) \prod_{j=0}^k u_j^{p_j} \\ = (1 + u_0)^{a-1} (1 + u_k)^{-b} \prod_{j=1}^{k-1} (1 + u_j)^{-1} \mathcal{E}_{a,b}(\mathbf{s}; L_1 - L_0, \dots, L_k - L_{k-1}). \end{aligned} \quad (2.12)$$

The series converges absolutely and locally normally on the full polydisc $|u_j| < 1$, uniformly on compact spectral and endpoint sets. Every fixed collection of independent spectral derivatives may be taken term by term.

Proof. For $|u| < 1$, $1 + u$ lies in the right half-plane, so $|\Im \log(1 + u)| < \pi/2$. Every consecutive sum of the t_j telescopes to $L_q - L_{p-1}$ and has imaginary part of absolute value less than π . Thus the right side of (2.12) is holomorphic on the entire stated polydisc, independently of k .

To identify its Taylor coefficients, use (2.10) locally and formally replace $\mathcal{H}_{a,b}(\mathbf{s} - \boldsymbol{\nu})$ by $\mathbf{y}^\nu$. The resulting expression is

$$(1 + u_0)^{a-y_1-1} \prod_{j=1}^{k-1} (1 + u_j)^{y_j - y_{j+1} - 1} (1 + u_k)^{y_k - b}.$$

Its coefficient of $\prod u_j^{p_j}$ is

$$\binom{a - y_1 - 1}{p_0} \prod_{j=1}^{k-1} \binom{y_j - y_{j+1} - 1}{p_j} \binom{y_k - b}{p_k}.$$

This is exactly the canonical strict-gap polynomial of the zero blocks. The analytic gap-deletion theorem in [2], with the independent-order continuation fixed in [3], identifies its finite polynomial expansion with the left coefficient in (2.12). Each coefficient uses only finitely many terms of (2.10); no interchange of a divergent formal order-shift sum is involved. Holomorphy of the right side now proves absolute local convergence of its Taylor series throughout the unit polydisc. Spectral derivatives obey the same Cauchy estimates uniformly on compact spectral sets. $\square$

Corollary 2.5 (Insertion sums and normalized primitives). *For every integer $r \geq 0$,*

$$\sum_{p_0 + \dots + p_k = r} \mathcal{H}_{a,b}(\{0\}^{p_0}, s_1, \dots, s_k, \{0\}^{p_k}) = \binom{a - b - k}{r} \mathcal{H}_{a,b}(s_1, \dots, s_k), \quad (2.13)$$

where the zero blocks between consecutive displayed free orders are included in the summand as in (2.12). Moreover,

$$\int_{\tau_0}^{t_j} \mathcal{E}_{a,b}(\mathbf{s}; t_1, \dots, \tau, \dots, t_k) d\tau = \mathcal{E}_{a,b}(\mathbf{s} + \mathbf{e}_j; \mathbf{t}) - \mathcal{E}_{a,b}(\mathbf{s} + \mathbf{e}_j; t_1, \dots, \tau_0, \dots, t_k), \quad (2.14)$$

along any path in the indicated slice of Ω_k . For $\mathbf{m} \in \mathbb{N}_0^k$,

$$\mathcal{E}_{a,b}(-\mathbf{m}; \mathbf{t}) = (-1)^{|\mathbf{m}|} \partial_{\mathbf{x}}^{\mathbf{m}} \mathcal{K}_{a,b}^{(k)}(-\mathbf{t}). \quad (2.15)$$

Proof. Set every $u_j = u$ in (2.12). All twists vanish and the prefactor becomes $(1 + u)^{a-b-k}$. The primitive is (2.9) with the additive normalization retained. For (2.15), use (2.11) with $N_j = m_j + 1$ and integrate the derivatives successively. Exponential decay removes the upper boundary terms and supplies precisely one additional minus sign for each integration. $\square$

At $a = b + N$ with $N \geq 0$ integral, the kernel is the finite ordered exponential sum and $\mathcal{E}_{a,b}$ is its weighted finite Dirichlet sum. The generating function is consequently a polynomial of total degree at most $N - k$ if $N \geq k$, and is zero if $N < k$. For $k = 1$ the theorem specializes to the previously proved single-Lerch generator. For $k > 1$ it resolves the requested joint convergence and simultaneous branch-cancellation problem. It supplies an explicit Gamma-separated kernel, with no claim that this kernel is minimal in an unspecified class.

3 Transverse index derivatives and Gamma–Stieltjes gaps

A formula proved only after setting an index to $-m$ cannot be differentiated in that index. The derivative sees information that a Bernoulli polynomial does not contain. This section constructs that information explicitly. We first obtain identities valid with an independent moving index, and only then differentiate and specialize. The kernel of Section 2 supplies a second, convergent integral realization of the same derivatives.

3.1 The joint Hurwitz completion

Put

$$Q(u, x) = \zeta(u, x) - \frac{1}{u-1}, \quad (3.1)$$

with its removable value at $u = 1$. This is entire in u and holomorphic for $\Re x > 0$. At the resonant point,

$$\partial_u^r Q(u, x)|_{u=1} = (-1)^r \gamma_r(x). \quad (3.2)$$

For $m, r \geq 0$ set

$$\ell_{m,r}(x) = \partial_u^r \zeta(u, x)|_{u=-m}. \quad (3.3)$$

The derivatives of Q at $-m$ differ from $\ell_{m,r}$ only by the constant $r!/(m+1)^{r+1}$. Thus they give the same gap differences.

For $\Re s$ sufficiently large, uniformly on compact sets of u , define the paired sum

$$\tilde{D}_{a,b}(s, u) = \sum_{n \geq 0} \{ (n+b)^{-s} Q(u, n+b) - (n+a)^{-s} Q(u, n+a) \}. \quad (3.4)$$

We initially require convergence of the separate one-sided series, not merely an accidental cancellation at an isolated exponent.

Theorem 3.1 (A jointly entire weighted-tail difference). *The continuation of (3.4) is the entire function*

$$\boxed{\tilde{D}_{a,b}(s, u) = Q(u, b)h_{a,b}(s) - \mathcal{H}_{a,b}(s, u)}. \quad (3.5)$$

Consequently the series

$$D_{m,r}(s; a, b) = \sum_{n \geq 0} \{ (n+b)^{-s} \ell_{m,r}(n+b) - (n+a)^{-s} \ell_{m,r}(n+a) \}, \quad \Re s > m+2, \quad (3.6)$$

has the entire continuation

$$D_{m,r}(s; a, b) = \ell_{m,r}(b)h_{a,b}(s) - \partial_u^r \mathcal{H}_{a,b}(s, u)|_{u=-m}. \quad (3.7)$$

For each $r \geq 0$, the similarly continued Stieltjes-weighted difference is

$$\begin{aligned} \sum_{n \geq 0} \left\{ \frac{\gamma_r(n+b)}{(n+b)^s} - \frac{\gamma_r(n+a)}{(n+a)^s} \right\} \\ = \gamma_r(b)h_{a,b}(s) - (-1)^r \partial_u^r \mathcal{H}_{a,b}(s, u)|_{u=1}. \end{aligned} \quad (3.8)$$

The last series is first interpreted, for example, on $\Re s > 1$.

Proof. In a domain of absolute convergence, summing the inner index gives

$$Z_c(s, u) = \zeta(s, c)\zeta(u, c) - \sum_{n \geq 0} (n+c)^{-s} \zeta(u, n+c).$$

Insert this identity into

$$\mathcal{H}_{a,b}(s, u) = Z_b(s, u) - Z_a(s, u) - \zeta(s, a)h_{a,b}(u).$$

The endpoint products cancel to leave

$$\sum_{n \geq 0} \{ (n+b)^{-s} \zeta(u, n+b) - (n+a)^{-s} \zeta(u, n+a) \} = \zeta(u, b)h_{a,b}(s) - \mathcal{H}_{a,b}(s, u).$$

Subtract $h_{a,b}(s)/(u-1)$ to obtain (3.5). Its right side is jointly entire by Theorem 2.3.

The Hurwitz asymptotic expansion, uniformly differentiated on a compact neighborhood of $u = -m$, gives $\ell_{m,r}(n+c) = O(n^{m+1}(1+\log^r n))$ on compact endpoint sets. A slightly enlarged compact set gives termwise spectral differentiation in a sufficiently far right half-plane; continuation then proves (3.7). Formula (3.2) gives (3.8). At $u = 1$, the coefficient $\gamma_r(n+c)$ is $O(1+\log^{r+1} n)$ on compact endpoint sets, proving the stated initial half-plane. $\square$

The whole function (3.5), rather than two separately regularized tails, fixes all constants. In particular the theorem covers simultaneous spectral derivatives at $s = 1, u = 1$.

3.2 A finite gap identity before specialization

For lattice functions f, g with enough decay, define

$$\begin{aligned} Z_c[f] &= \sum_{n \geq 0} f(n+c), & Z_c[f, g] &= \sum_{n > j \geq 0} f(n+c)g(j+c), \\ \mathcal{H}_{a,b}[f, g] &= Z_b[f, g] - Z_a[f, g] - Z_a[f]\{Z_b[g] - Z_a[g]\}. \end{aligned} \quad (3.9)$$

For longer decorated words, define

$$Z_c[f_1, \dots, f_k] = \sum_{n_1 > \dots > n_k \geq 0} \prod_{j=1}^k f_j(n_j + c),$$

and define $\mathcal{H}_{a,b}[f_1, \dots, f_k]$ by the same prefix-suffix recursion as (1.6); the displayed formula (3.9) is its depth-two case. All decorated expressions below mean these initial definitions and their specified continuations. They do not mean arbitrary interpolations of finite sums. Write $f_s(x) = x^{-s}$ and use pointwise products of letters.

Theorem 3.2 (The moving-index gap and every transverse derivative). *The two functions*

$$A(s, u, t) = \mathcal{H}_{a,b}[f_s Q(u, \cdot), f_t], \quad B(s, u, t) = \mathcal{H}_{a,b}[f_s, f_t Q(u, \cdot + 1)]$$

continue individually to entire functions of (s, u, t) . They obey

$$A(s, u, t) = \mathcal{H}_{a,b}(u, s, t) + \mathcal{H}_{a,b}(s+u, t) + Q(u, a)\mathcal{H}_{a,b}(s, t), \quad (3.10)$$

$$B(s, u, t) = A(s, u, t) + \mathcal{H}_{a,b}(s, u, t). \quad (3.11)$$

Thus

$$\boxed{\mathcal{H}_{a,b}(s, u, t) = \mathcal{H}_{a,b}[f_s, f_t Q(u, \cdot + 1)] - \mathcal{H}_{a,b}[f_s Q(u, \cdot), f_t]}. \quad (3.12)$$

For $m, r \geq 0$ this gives

$$\begin{aligned} J_{m,r}(s, t; a, b) &:= \partial_u^r \mathcal{H}_{a,b}(s, u, t)|_{u=-m} \\ &= \mathcal{H}_{a,b}[f_s, f_t \ell_{m,r}(\cdot + 1)] - \mathcal{H}_{a,b}[f_s \ell_{m,r}(\cdot), f_t]. \end{aligned} \quad (3.13)$$

In particular this is an identity for arbitrary independent s, t and arbitrary right-half-plane endpoints.

Proof. First replace Q by ζ and take all real parts large. In the decorated sum $Z_c[f_s \zeta(u, \cdot), f_t]$, the additional tail index is either larger than the outer index or equal to it. Accordingly

$$Z_c[f_s \zeta(u, \cdot), f_t] = Z_c(u, s, t) + Z_c(s+u, t), \quad Z_c[f_s \zeta(u, \cdot)] = Z_c(u, s) + \zeta(s+u, c).$$

Substitute these identities in (3.9). Expanding the relative recursion (1.6) and collecting the same cuts gives

$$\mathcal{H}_{a,b}[f_s \zeta(u, \cdot), f_t] = \mathcal{H}_{a,b}(u, s, t) + \mathcal{H}_{a,b}(s + u, t) + \zeta(u, a) \mathcal{H}_{a,b}(s, t).$$

In the other decorated double, its two indices above the lower one can be ordered in either direction or can be equal. These three cases give $Z_c(s, u, t)$, $Z_c(u, s, t)$, and $Z_c(s + u, t)$. The same relative recursion gives the extra term $\mathcal{H}_{a,b}(s, u, t)$.

Subtracting the constant letter $1/(u - 1)$ proves (3.10)–(3.11). Their right sides are entire, so they also provide individual continuations of the decorated doubles through every spectral intersection. Their difference proves (3.12). Differentiate this entire identity. At $u = -m$ the constants that distinguish the derivatives of Q from $\ell_{m,r}$ cancel between the two doubles. This proves (3.13). $\square$

The arbitrary-depth version of the counting argument will be useful when several unmarked slots have first been removed. It also makes the general-endpoint meaning of every boundary gap explicit.

Lemma 3.3 (A decorated word with one moving tail). *Let $\mathbf{s} = (s_1, \dots, s_k)$ and $1 \leq i \leq k$. Then the decorated relative word has the jointly entire continuation*

$$\begin{aligned} & \mathcal{H}_{a,b}[f_{s_1}, \dots, f_{s_i} Q(u, \cdot), \dots, f_{s_k}] \\ &= Q(u, a) \mathcal{H}_{a,b}(s_1, \dots, s_k) + \sum_{j=0}^{i-1} \mathcal{H}_{a,b}(s_1, \dots, s_j, u, s_{j+1}, \dots, s_k) \\ & \quad + \sum_{j=1}^i \mathcal{H}_{a,b}(s_1, \dots, s_j + u, \dots, s_k). \end{aligned} \tag{3.14}$$

In particular the top and bottom boundary identities are

$$\mathcal{H}_{a,b}(u, \mathbf{s}) = \mathcal{H}_{a,b}[f_{s_1} Q(u, \cdot + 1), f_{s_2}, \dots, f_{s_k}] - Q(u, a) \mathcal{H}_{a,b}(\mathbf{s}), \tag{3.15}$$

$$\mathcal{H}_{a,b}(\mathbf{s}, u) = Q(u, b) \mathcal{H}_{a,b}(\mathbf{s}) - \mathcal{H}_{a,b}[f_{s_1}, \dots, f_{s_k} Q(u, \cdot)]. \tag{3.16}$$

Proof. First replace $Q(u, x)$ by $\zeta(u, x)$ and take all real parts large. In the infinite ordered sum, the extra tail index is at least the index in position i . Its possible strict positions are the i slots preceding that position, and its possible equalities are the first i existing indices. Thus the infinite decorated sum is exactly the sum of the insertion terms and merger terms in (3.14), with Z_c in place of $\mathcal{H}$ and without the endpoint term.

Expand each such Z_b by deconcatenation and group terms by the cut in the original undecorated word. For a cut before position i , insertions and mergers lying wholly in the prefix give $\zeta(u, a) Z_a(\text{prefix})$ by the ordinary singleton stuffle identity. The remaining insertions and mergers lie in the suffix. For a cut at or after position i , the same counting identity gives $Z_a(\text{decorated prefix})$. These are exactly the two types of cuts in the defining recursion for the decorated relative word. Triangular uniqueness proves the claimed identity with ζ ; subtracting $\mathcal{H}(\mathbf{s})/(u - 1)$ replaces its endpoint value by $Q(u, a)$ and proves (3.14). Its right side provides the jointly entire continuation.

At $i = 1$, use $Q(u, x + 1) = Q(u, x) - x^{-u}$ to obtain (3.15). At $i = k$, the relative singleton stuffle identity combines all insertions and mergers, and $Q(u, b) - Q(u, a) = h_{a,b}(u)$ gives (3.16). The difference of the decorations at adjacent positions $i + 1$ and i , using $Q(u, \cdot + 1)$ at the lower position, leaves exactly the insertion at the gap between them. This proves the interior gap identity at any depth as well. No integer-endpoint uniqueness principle is used. $\square$

At an integer interval $a = b + N$, the meaning of the gap is especially transparent:

$$J_{m,r}(s, t; b + N, b) = \sum_{N > i > k \geq 0} \frac{\ell_{m,r}(k + b + 1) - \ell_{m,r}(i + b)}{(i + b)^s (k + b)^t}. \quad (3.17)$$

Indeed the middle-index sum is

$$\sum_{k < j < i} (j + b)^m (-\log(j + b))^r = \ell_{m,r}(k + b + 1) - \ell_{m,r}(i + b).$$

The proof above establishes the general-endpoint identity independently of this finite illustration.

The Hurwitz shift identity also gives an equivalent form using only unshifted decorated letters:

$$J_{m,r}(s, t; a, b) = \mathcal{H}_{a,b}[f_s, f_t \ell_{m,r}] - \mathcal{H}_{a,b}[f_s \ell_{m,r}, f_t] - \partial_t^r \mathcal{H}_{a,b}(s, t - m). \quad (3.18)$$

The last term is essential. For $m = r = 0$, it recovers exactly

$$\mathcal{H}(s, 0, t) = \mathcal{H}(s - 1, t) - \mathcal{H}(s, t - 1) - \mathcal{H}(s, t),$$

the strict zero-gap deletion rule.

3.3 Gamma gaps, Stieltjes primitives, and repeated primitives

For the first transverse derivative at zero, (1.5) turns (3.17) into

$$\partial_u \mathcal{H}_{b+N,b}(s, u, t)|_{u=0} = \sum_{N > i > k \geq 0} \frac{\log \Gamma(k + b + 1) - \log \Gamma(i + b)}{(i + b)^s (k + b)^t}. \quad (3.19)$$

The constants $\log(2\pi)/2$ disappear. This is the requested transverse identity in explicit Gamma coordinates, with (3.13) specifying its general-endpoint counterpart.

Differentiating $-u\zeta(1 + u, x)$ at $u = 0$ gives, for $r \geq 1$,

$$\frac{d}{dx} \ell_{0,r}(x) = (-1)^r r \gamma_{r-1}(x). \quad (3.20)$$

Therefore

$$\ell_{0,r}(z + 1) - \ell_{0,r}(y) = (-1)^{r+1} r \int_{z+1}^y \gamma_{r-1}(x) dx. \quad (3.21)$$

The path lies in the right half-plane. The integral is path independent, and its additive normalization is fixed by the displayed endpoints. Combining (3.13) with (3.21) gives all the corresponding harmonic identities with Stieltjes-primitive gaps. At $u = 1$, the same entire identity instead gives

$$\partial_u^r \mathcal{H}(s, u, t)|_{u=1} = (-1)^r \{ \mathcal{H}[f_s, f_t \gamma_r(\cdot + 1)] - \mathcal{H}[f_s \gamma_r, f_t] \}. \quad (3.22)$$

These statements retain the ordered information; they do not claim a reduction of every resulting value to depth-one constants.

A standard balanced normalization packages repeated primitives of log-Gamma [13]:

$$\mathcal{B}_m(x) = \frac{\zeta'(-m, x) + H_m \zeta(-m, x)}{m!}, \quad H_0 = 0. \quad (3.23)$$

Here $\mathcal{B}_0(x) = \log \Gamma(x) - \frac{1}{2} \log(2\pi)$, and

$$\mathcal{B}'_m(x) = \mathcal{B}_{m-1}(x) \quad (m \geq 1). \quad (3.24)$$

To check this, differentiate the Hurwitz identity $\partial_x \zeta(u, x) = -u \zeta(u + 1, x)$ once in u , then use $H_m = H_{m-1} + 1/m$. The factor $m!$ in (3.23) is exactly the one needed for (3.24).

It follows that the repeated-Gamma-primitive gap has the exact identity

$$\begin{aligned} & \mathcal{H}[f_s, f_t \mathcal{B}_m(\cdot + 1)] - \mathcal{H}[f_s \mathcal{B}_m, f_t] \\ &= \frac{J_{m,1}(s, t) + H_m \mathcal{H}(s, -m, t)}{m!}. \end{aligned} \quad (3.25)$$

The second term is a finite Bernoulli reduction. Thus no undetermined polynomial or integration constant is hidden in the primitive ladder.

As one elementary Stieltjes check, let $h = h_{a,b}$. Differentiating the stuffle of a singleton with $\mathcal{H}(s, t)$ gives

$$\begin{aligned} & \partial_u \{ \mathcal{H}(u, s, t) + \mathcal{H}(s, u, t) + \mathcal{H}(s, t, u) \} |_{u=0} \\ &= h'(0) \mathcal{H}(s, t) - (\partial_s + \partial_t) \mathcal{H}(s, t). \end{aligned} \quad (3.26)$$

At $s = t = 1$, use $\mathcal{H}(v, v) = (h(v)^2 - h(2v))/2$ to obtain the fully specified expression

$$\frac{h'(0)}{2} \{ h(1)^2 - h(2) \} + h(1) \{ \gamma_1(b) - \gamma_1(a) \} + \zeta'(2, b) - \zeta'(2, a). \quad (3.27)$$

This is a consistency specialization of the classical stuffle algebra, not a separate claim of a new product principle.

3.4 Absolutely convergent transverse Mellin formulas

The following elementary normalized-Mellin lemma makes the transverse construction directly computable.

Lemma 3.4 (Every derivative at a nonpositive Mellin order). *Suppose f is smooth on $[0, \infty)$, extends holomorphically near zero, and all its derivatives decrease exponentially at infinity. Let*

$$\mathcal{M}_f(u) = \frac{1}{\Gamma(u)} \int_0^\infty y^{u-1} f(y) dy, \quad \Re u > 0.$$

It has an entire continuation. Put $c_j = [v^j] \Gamma(v)^{-1}$. For $m \geq 0$,

$$\mathcal{M}_f(-m) = (-1)^m f^{(m)}(0), \quad (3.28)$$

$$\mathcal{M}_f^{(r)}(-m) = (-1)^m r! \sum_{j=1}^r \frac{c_j}{(r-j)!} \int_0^\infty \frac{\log^{r-j} y}{y} \{ f^{(m)}(y) - f^{(m)}(0) e^{-y} \} dy, \quad r \geq 1. \quad (3.29)$$

All integrals in (3.29) are ordinary absolutely convergent integrals.

Proof. Repeated integration by parts gives $\mathcal{M}_f(u) = (-1)^m \mathcal{M}_{f^{(m)}}(u + m)$, initially on $\Re u > 0$ and then by continuation. It is enough to expand at zero. Write $g = f^{(m)}$ and use the exact identity

$$\mathcal{M}_g(v) = g(0) + \frac{1}{\Gamma(v)} \int_0^\infty y^{v-1} \{ g(y) - g(0) e^{-y} \} dy.$$

The last integral is holomorphic for $\Re v > -1$, locally at zero. Its q th derivative at zero is the integral with $\log^q y/y$. The bracket is $O(y)$ near zero and decreases exponentially at infinity. Coefficient multiplication by $\Gamma(v)^{-1}$ proves (3.29). The continuation and value formula also follow from repeated integration by parts with an arbitrarily large number of derivatives. $\square$

The coefficients c_j are explicit Gamma Taylor coefficients:

$$c_1 = 1, \quad c_2 = \gamma, \quad c_3 = \frac{\gamma^2 - \zeta(2)}{2},$$

and in every degree they follow from

$$\frac{1}{\Gamma(v)} = v \exp \left(\gamma v + \sum_{q \geq 2} \frac{(-1)^{q+1} \zeta(q)}{q} v^q \right).$$

Theorem 3.5 (Convergent formula for every transverse gap). *Let $\Re s, \Re t > 0$, $m \geq 0$, and $r \geq 1$. In (3.29), set*

$$f(y) = \mathcal{K}_{a,b}^{(3)}(x, y, z).$$

Then $J_{m,r}(s, t; a, b)$ is the normalized (x, z) Mellin transform of its right side. Explicitly it is

$$\begin{aligned} & \frac{(-1)^m r!}{\Gamma(s) \Gamma(t)} \sum_{j=1}^r \frac{c_j}{(r-j)!} \int_0^\infty \int_0^\infty x^{s-1} z^{t-1} \\ & \cdot \int_0^\infty \frac{\log^{r-j} y}{y} \left\{ \partial_y^m \mathcal{K}_{a,b}^{(3)}(x, y, z) - \partial_y^m \mathcal{K}_{a,b}^{(3)}(x, 0, z) e^{-y} \right\} dy dz dx. \end{aligned} \quad (3.30)$$

All these integrals are absolutely convergent. Their common continuation is entire in s, t . Every further fixed spectral derivative can be taken in the convergent integrals on compact subsets of $\Re s, \Re t > 0$.

Proof. Lemma 2.1 gives uniform exponential control of all derivatives of the kernel, including at $y = 0$. The inner bracket is uniformly $O(y)e^{-\delta(x+z)}$ as $y \downarrow 0$; for $y \geq 1$ it is bounded by $Ce^{-\delta(x+z)}(e^{-\delta y} + e^{-y})$. Consequently the inner logarithmic integrals and the two outer Mellin integrals are absolutely convergent. Fubini's theorem and the preceding lemma give the derivative of $\mathcal{E}_{a,b}(s, u, t; \mathbf{0}) = \mathcal{H}_{a,b}(s, u, t)$. Theorem 2.3 gives the entire continuation. Powers of $\log x$ and $\log z$ from further spectral derivatives preserve the same integrable majorants on compact half-plane subsets. $\square$

For $m = 0, r = 1$ the formula contains just

$$\int_0^\infty \frac{\mathcal{K}^{(3)}(x, y, z) - \mathcal{K}^{(3)}(x, 0, z) e^{-y}}{y} dy.$$

This supplies a convergent realization of the log-Gamma gap at arbitrary noninteger endpoints, without separate divergent ordered sums.

3.5 All negative outer values and fixed Barnes constants

The weighted difference in Theorem 3.1 has especially explicit values at nonpositive outer orders.

Theorem 3.6 (Finite Bernoulli formula for the transverse weighted tails). *Let $d, m, r \geq 0$ and $q = d+1$. Then*

$$\begin{aligned} D_{m,r}(-d; a, b) &= \frac{1}{q} \left\{ B_q(a) \ell_{m,r}(a) - B_q(b) \ell_{m,r}(b) \right. \\ & \quad \left. + \sum_{k=0}^q \binom{q}{k} B_{q-k}(1) h_{a,b}^{(r)}(-m-k) \right\}. \end{aligned} \quad (3.31)$$

Every term is an endpoint Hurwitz derivative or a Bernoulli polynomial.

Proof. The required top-index deletion identity is

$$\mathcal{H}(-d, u) = \frac{1}{q} \left\{ B_q(a)h(u) - \sum_{k=0}^q \binom{q}{k} B_{q-k}(1)h(u-k) \right\}. \quad (3.32)$$

For a direct analytic derivation, sum the leading index of $Z_c(v, u)$ as $\zeta(v, n+c+1)$, then continue v to $-d$ while $\Re u$ is large. Use $\zeta(-d, x+1) = -B_q(x+1)/q$ and expand the Bernoulli polynomial. Insert the result in the relative quotient. The leading singleton $Z_a(-d) = -B_q(a)/q$ contributes the first term. This proves (3.32) on a nonempty domain, and entireness in u extends it.

Differentiate (3.32) r times in u and set $u = -m$. Substitute in (3.7) and use $h(-d) = (B_q(a) - B_q(b))/q$. The terms involving $B_q(a)\ell_{m,r}(b)$ cancel, giving (3.31). $\square$

For $m = 0, r = 1, d = 0$, the standard Barnes relation [14, 12] gives the particularly simple identity

$$D_{0,1}(0; a, b) = \log G(a) - \log G(b) - \frac{a-b}{2} \log(2\pi). \quad (3.33)$$

Here G is Barnes' function with its standard asymptotic normalization, $G(1) = 1$, and $G(x+1) = \Gamma(x)G(x)$. Holomorphic logarithms in the right half-plane are normalized from the positive real axis. Indeed the $q = 1$ formula is

$$(a-1)\ell_{0,1}(a) - (b-1)\ell_{0,1}(b) + \zeta'(-1, b) - \zeta'(-1, a),$$

and $\log G(x) = (x-1)\log \Gamma(x) + \zeta'(-1) - \zeta'(-1, x)$ proves (3.33). At an integer endpoint difference it also agrees with the finite product obtained by iterating the functional equation. General d gives the analogous finite endpoint formula in higher Hurwitz derivatives, without choosing an unspecified multiple Gamma normalization.

3.6 A finite family after arbitrary unmarked deletions

The preceding construction is stable under inserting any finite word of unmarked nonpositive indices. We state exactly the scope.

Proposition 3.7 (One marked index with arbitrary nonpositive decoration). *Suppose a word contains $k \geq 1$ independent free indices, one marked index u , and unmarked indices $-m_1, \dots, -m_e$, with $m_i \geq 0$. Put $M = \sum_i (m_i + 1)$. After differentiating the marked index r times and setting $u = -m$, the word is a finite polynomial-endpoint combination of k -letter relative sums whose letters are ordinary shifted powers, ordinary spectral derivatives of those powers, and at most one factor*

$$\ell_{m+\nu, r}(x), \quad 0 \leq \nu \leq M.$$

Endpoint values of the same finite family may occur. Thus the new gap functions needed for this one-marked problem are a specified finite list. This is a bound for this representation, not an assertion of minimal arithmetic depth.

Proof. First regard u as free. The finite Bernoulli deletion rules for each unmarked $-m_i$ hold identically in all remaining indices: sum its finite strict gap by a Bernoulli polynomial, or use the leading-tail argument of (3.32) at the upper boundary, and insert the result in the relative recursion. An interior deletion has the two endpoint terms

$$\frac{1}{m_i + 1} \sum_{j=0}^{m_i+1} \binom{m_i+1}{j} \{ B_{m_i+1-j} \mathcal{H}(\dots, s-j, t, \dots) - B_{m_i+1-j}(1) \mathcal{H}(\dots, s, t-j, \dots) \}.$$

It shifts adjacent surviving orders down by nonnegative integers. The total shift budget is at most M , even when adjacent unmarked indices are combined during the deletion. Coefficients are endpoint polynomials and are independent of every surviving free order.

Now differentiate in u and set $u = -m$. Each surviving marked order is $-m - \nu$ for some $0 \leq \nu \leq M$. Apply Lemma 3.3 at the surviving marked position. Its adjacent-decoration difference gives the interior gap, and (3.15)–(3.16) give the boundary gaps. Their finite-interval forms are $\ell(y+1) - \ell(a)$ at the top and $\ell(b) - \ell(y)$ at the bottom. Only one decorated factor is introduced. Finally $\ell_{m+\nu,r}(x+1) = \ell_{m+\nu,r}(x) - x^{m+\nu}(-\log x)^r$ removes shifted potentials in favor of ordinary free-order derivatives. The analytic identities at every step were proved with the marked order still free, so the transverse differentiation is legitimate. $\square$

There is also an analytic all-decoration version. Take any fixed free word with one marked slot, apply Theorem 2.4, then differentiate that slot at $-m$. Local normality proves convergence on the same full unit polydisc. Formula (3.29), applied to the marked kernel coordinate, gives the corresponding logarithmic-integral generator. This simultaneously extends R7 and R8 to every derivative order and every nonpositive marked center.

If there are no other free indices, the remaining singleton is $h_{a,b}^{(r)}(-m - \nu)$, so no decorated lower-depth sum is needed. Several independently marked indices are a further problem: this proposition makes no claim that their repeated gap primitives close in the same one-marked list.

4 Mixed centered harmonic-tail moments

The source reports evaluate every centered even moment of a trigamma square [4, 2]; the harmonic-parity report asks for the corresponding evaluation for two unequal even-order harmonic tails. We resolve that two-factor problem for all orders and all moments. The resulting finite formula reduces entirely to ordinary zeta values and their products. We then package the infinite moment sequence into one analytic generator with explicit polylogarithmic primitives. The centered parity mechanism is inherited from the source criterion; the evaluation and the generator are the additional results here. Euler's classical odd-weight reduction supplies the arithmetic step [6].

4.1 Continuation and ordinary convergent representatives

Let $a, b \geq 2$ be even integers, and put

$$T_a(n) = \zeta(a, n+1) = \zeta(a) - H_n^{(a)} = \frac{\psi^{(a-1)}(n+1)}{(a-1)!}, \quad x_n = n + \frac{1}{2}. \quad (4.1)$$

The last equality uses that a is even. Define

$$F_{a,b}(s) = \sum_{n=0}^{\infty} \frac{T_a(n)T_b(n)}{x_n^s}, \quad V_{a,b}(M) = F_{a,b}(-2M), \quad M \geq 0. \quad (4.2)$$

The series for $F_{a,b}$ converges absolutely when $\Re s > 3 - a - b$. The value at $-2M$ is understood through the continuation proved below whenever the unmodified series diverges.

Write $\alpha = a + b - 2$, and define rational coefficients

$$t_{a,0} = \frac{1}{a-1}, \quad t_{a,k} = \frac{B_{2k}(1/2)}{(2k)!} (a)_{2k-1} \quad (k \geq 1), \quad c_k = \sum_{j=0}^k t_{a,j} t_{b,k-j}. \quad (4.3)$$

Here $(a)_r$ is the rising factorial, $B_j(x)$ is the Bernoulli polynomial, and $B_j = B_j(0)$, with $B_1 = -1/2$. The centered Hurwitz expansions [7, §25.11(xii)] give

$$T_a(n) \sim \sum_{k \geq 0} t_{a,k} x_n^{1-a-2k}, \quad T_a(n)T_b(n) \sim \sum_{k \geq 0} c_k x_n^{-\alpha-2k}. \quad (4.4)$$

Only finite truncations of these asymptotic expansions are used.

Proposition 4.1 (Regularity and bracketed moments). *The function $F_{a,b}$ extends meromorphically to the complex plane. Its only possible poles are simple poles at $s = 3 - a - b - 2k$, $k \geq 0$, with residue c_k . In particular every even spectral integer is regular. For*

$$P_M(x) = \sum_{\alpha+2k \leq 2M} c_k x^{2M-\alpha-2k}, \quad (4.5)$$

one has the ordinary convergent identity

$$V_{a,b}(M) = \sum_{n=0}^{\infty} \left[x_n^{2M} T_a(n) T_b(n) - P_M(x_n) \right]. \quad (4.6)$$

The polynomial is zero if its indexing set is empty. The bracket is $O(x_n^{-2})$.

Proof. For any fixed K , subtract the first $K+1$ terms of (4.4) inside the defining series. The remainder converges normally in a right half-plane extending farther to the left, while the removed terms sum to $\sum_{k=0}^K c_k \zeta(s + \alpha + 2k, 1/2)$. Increasing K provides compatible continuations and the stated poles and residues. Since $a + b$ is even, all these potential poles are odd integers. At $s = -2M$, every nondecaying term has a nonnegative even power of x_n . Its continued sum is $\zeta(-2j, 1/2) = 0$, including $j = 0$. After their removal the next possible power is x_n^{-2} , which proves (4.6). $\square$

Corollary 4.2 (Every spectral derivative with its correction terms). *For $r, M \geq 0$, every derivative at a centered even spectral value has the ordinary convergent representation*

$$\begin{aligned} F_{a,b}^{(r)}(-2M) &= \sum_{n=0}^{\infty} (-\log x_n)^r \left[x_n^{2M} T_a(n) T_b(n) - P_M(x_n) \right] \\ &\quad + \sum_{\alpha+2k \leq 2M} c_k \zeta^{(r)}(\alpha + 2k - 2M, 1/2). \end{aligned} \quad (4.7)$$

The second sum vanishes for $r = 0$. It must in general be retained for $r \geq 1$.

Proof. Use the same finite subtractions as in (4.5) at a moving s in a neighborhood of $-2M$. The remainder series and all its fixed spectral derivatives converge locally normally there, since their terms are bounded by $C x_n^{-2+\varepsilon} (1 + |\log x_n|^r)$ with some $\varepsilon < 1$. Differentiation of the finite Hurwitz correction gives exactly (4.7). $\square$

For example, the first derivative in the $(2, 4)$ family at $s = -4$ is

$$F'_{2,4}(-4) = - \sum_{n \geq 0} \log x_n \left[x_n^4 T_2(n) T_4(n) - \frac{1}{3} \right] - \frac{\log 2}{6}, \quad (4.8)$$

because $\zeta'(0, 1/2) = -\frac{1}{2} \log 2$. This identity fixes the logarithmic remainder and its constant; it does not claim that the remainder is an ordinary-zeta combination.

We will also use a concrete equivalent characterization. Let CT_N mean the coefficient of $N^0(\log N)^0$ in an asymptotic expansion in the integer cutoff N . Then

$$V_{a,b}(M) = \text{CT}_N \sum_{n=0}^{N-1} x_n^{2M} T_a(n) T_b(n). \quad (4.9)$$

Indeed the partial sum of each removed even polynomial power is $B_{2j+1}(N + 1/2)/(2j + 1)$, an odd polynomial in N with zero constant term. The summable remainder in (4.6) tends to its ordinary sum. The coordinate N in (4.9) is part of the statement; no change of endpoint coordinate is implicit.

4.2 A finite formula at every moment order

The double-zeta convention used below is

$$\zeta(u, v) = \sum_{j > k \geq 1} \frac{1}{j^u k^v}.$$

Thus the first exponent belongs to the larger summation variable. For each positive odd integer d , define

$$K_{a,b}(d) = \begin{cases} \zeta(b, a-d) + \frac{1}{2}\zeta(a+b-d), & d < a, \\ \sum_{j=1}^{\min(b/2, (d-a+1)/2)} \frac{\binom{d-a+1}{2j} B_{d-a+1-2j}}{d-a+1} \zeta(b-2j), & d > a, \end{cases} \quad (4.10)$$

and put

$$E_{a,b}(d) = K_{a,b}(d) + K_{b,a}(d). \quad (4.11)$$

There is no missing case $d = a$: d is odd and a even. All double zetas in the first branch are ordinary convergent sums. In the second branch, $\zeta(0) = -1/2$ has its usual continued meaning.

Theorem 4.3 (All mixed even-tail moments). *For even $a, b \geq 2$ and every integer $M \geq 0$,*

$$V_{a,b}(M) = \frac{1}{2M+1} \sum_{\ell=0}^M \binom{2M+1}{2\ell+1} B_{2M-2\ell}(1/2) E_{a,b}(2\ell+1). \quad (4.12)$$

The summation and every coefficient in this formula are finite and explicit.

The $j = b/2$ term in (4.10), whenever its range includes it, records an exact rational boundary contribution through $\zeta(0) = -1/2$. Discarding this term would already give an incorrect answer for divergent centered moments of the trigamma square.

Proof. First form the exact polynomial partial sum

$$\begin{aligned} S_M(k) &= \sum_{n=0}^{k-1} (n+1/2)^{2M} = \frac{B_{2M+1}(k+1/2)}{2M+1} = \sum_{\ell=0}^M s_{M,\ell} k^{2\ell+1}, \\ s_{M,\ell} &= \frac{\binom{2M+1}{2\ell+1}}{2M+1} B_{2M-2\ell}(1/2). \end{aligned} \quad (4.13)$$

The tail recurrence gives

$$\begin{aligned} T_a(k-1)T_b(k-1) - T_a(k)T_b(k) \\ = k^{-a}T_b(k-1) + k^{-b}T_a(k-1) - k^{-a-b}. \end{aligned}$$

Applying finite summation by parts, without taking any divergent limits, therefore yields

$$\sum_{n=0}^{N-1} x_n^{2M} T_a(n) T_b(n) = \sum_{\ell=0}^M s_{M,\ell} \mathcal{E}_{2\ell+1}(N), \quad (4.14)$$

where

$$\mathcal{E}_d(N) = N^d T_a(N) T_b(N) + \sum_{k=1}^N k^{d-a} T_b(k-1) + \sum_{k=1}^N k^{d-b} T_a(k-1) - H_N^{(a+b-d)}. \quad (4.15)$$

We compute its cutoff constant explicitly.

If $d < a$, the first oriented sum in (4.15) is absolutely convergent, and direct rearrangement gives

$$\sum_{k=1}^{\infty} k^{d-a} T_b(k-1) = \zeta(b, a-d) + \zeta(a+b-d). \quad (4.16)$$

If $d > a$, put $p = d - a$, a positive odd integer. The Faulhaber polynomial is

$$Q_p(X) = \sum_{k=1}^X k^p = \frac{1}{2} X^p + \sum_{j=1}^{(p+1)/2} q_{p,2j} X^{2j}, \quad q_{p,2j} = \frac{\binom{p+1}{2j} B_{p+1-2j}}{p+1}. \quad (4.17)$$

The finite-sum notation specifies the polynomial by its values at nonnegative integers. Define $q_{p,h} = 0$ outside its displayed nonzero even degrees, whenever an even h is used below. A finite rearrangement which retains the infinite tail gives the exact equality

$$\sum_{k=1}^N k^p T_b(k-1) = \sum_{h \geq 1} [X^h] Q_p(X) H_N^{(b-h)} + Q_p(N) T_b(N). \quad (4.18)$$

For $v > 1$, the cutoff constant of $H_N^{(v)}$ is $\zeta(v)$; for $v = 1$ it is Euler's constant γ ; and for integer $v \leq 0$ it is zero because the finite power sum is a polynomial in N with zero constant term.

At the potential logarithmic collision $d = a + b - 1$, the half-leading monomial in each of the two Faulhaber polynomials contributes $\gamma/2$. The diagonal term in (4.15) contributes $-\gamma$. These cancel. In particular, this proof never evaluates an isolated $\zeta(1)$.

We now evaluate the constant contributions of the explicit tails. Write

$$T_a(N) \sim \sum_j \tau_a(j) N^{-j}, \quad \begin{cases} \tau_a(a-1) = 1/(a-1), \\ \tau_a(a) = -1/2, \\ \tau_a(a+2k-1) = B_{2k}(a)_{2k-1}/(2k)!, \quad k \geq 1, \end{cases} \quad (4.19)$$

with all other coefficients zero. Since a, b are even and d odd, an odd total degree in the product can only use one exceptional even index a or b . Hence

$$\text{CT}_N N^d T_a(N) T_b(N) = -\frac{1}{2} \tau_b(d-a) - \frac{1}{2} \tau_a(d-b). \quad (4.20)$$

For $p = d - a > 0$, the only odd monomial of Q_p is $X^p/2$, and the only even index in T_b is b . Therefore

$$\text{CT}_N Q_p(N) T_b(N) = \frac{1}{2} \tau_b(p) - \frac{1}{2} q_{p,b}. \quad (4.21)$$

The τ terms in (4.20) cancel those from the two instances of (4.21). The remaining constant is

$$-\frac{1}{2}(q_{d-a,b} + q_{d-b,a}), \quad (4.22)$$

where a term is zero if its first index is not positive. This is exactly the contribution of the $\zeta(0)$ terms in (4.10).

For completeness consider the remaining diagonal constant. If $W = a + b - d \geq 3$, a branch $d < a$ contributes the full $\zeta(W)$ in (4.16), and a branch $d > a$ contributes $\zeta(W)/2$ from the half-leading term in (4.18). After subtracting the diagonal $\zeta(W)$, exactly one half remains for each branch $d < a$. When $W = 1$ the constants cancel as already shown. When $W < 1$, the diagonal power sum has zero constant term. The other surviving Faulhaber terms have even degree and yield the even zeta values in (4.10). Thus

$$\text{CT}_N \mathcal{E}_d(N) = E_{a,b}(d).$$

Taking constants in (4.14) and applying (4.9) proves (4.12). $\square$

4.3 Reduction to ordinary zeta values

For clarity, the classical reduction needed to eliminate every double coordinate is written out. Let $m \geq 2$ be even, let $n \geq 1$ be odd, and put $w = m + n$ and $\mathcal{U}_{m,n} = \zeta(m, n) + \zeta(w)/2$. Euler's formulas, in the convention above, give

$$\mathcal{U}_{m,1} = \frac{m+1}{2}\zeta(m+1) - \sum_{j=1}^{m/2-1} \zeta(2j)\zeta(m+1-2j), \quad (4.23)$$

and, for $n \geq 3$,

$$\begin{aligned} \mathcal{U}_{m,n} = & \frac{1}{2} \binom{w}{m} \zeta(w) + \zeta(m)\zeta(n) \\ & - \sum_{j=1}^{(w-3)/2} \left[\binom{w-2j-1}{m-1} + \binom{w-2j-1}{n-1} \right] \zeta(2j)\zeta(w-2j). \end{aligned} \quad (4.24)$$

These are the ordinary specialization of the classical signed Euler reduction; an elementary proof from partial fractions is provided in [6, Eq. (2)–(3) and Proposition 1]. No divergent $\zeta(1)$ appears in (4.23)–(4.24). The first branch of (4.10) is exactly $\mathcal{U}_{b,a-d}$, so the formulas apply to every such branch.

Corollary 4.4 (Ordinary-zeta closure and a fixed spanning list). *Every $V_{a,b}(M)$ is a rational linear combination of 1, ordinary even zeta values, ordinary odd zeta values, and products of one even and one odd zeta value. Put $B = \max(a, b)$. The entire moment sequence lies in the rational span of the at most B coordinates*

$$E_{a,b}(1), E_{a,b}(3), \dots, E_{a,b}(B-1), \quad 1, \zeta(2), \zeta(4), \dots, \zeta(B-2). \quad (4.25)$$

The positive even-zeta sublist is empty when $B = 2$.

Proof. Ordinary-zeta closure follows by inserting (4.23)–(4.24) into (4.10). For odd $d > B$, both branches in (4.10) are finite combinations of $\zeta(0), \zeta(2), \dots, \zeta(B-2)$. Formula (4.12) then proves the spanning assertion. $\square$

The corollary is a finite spanning theorem, not a claim that its coordinates are arithmetically independent or minimal. It answers the ordinary-constant reduction question for every pair of even-order tails; increasing the moment order introduces no additional constants.

4.4 The first unequal-order family

For $a = 2$ and $b = 4$, the first five coordinates are

$$\begin{aligned} E_{2,4}(1) &= \frac{15}{2}\zeta(5) - 3\zeta(2)\zeta(3), & E_{2,4}(3) &= \frac{1}{2}\zeta(2) + \frac{3}{2}\zeta(3), \\ E_{2,4}(5) &= \frac{1}{4}\zeta(2) - \frac{3}{8}, & E_{2,4}(7) &= -\frac{1}{12}\zeta(2) - \frac{1}{3}, \\ E_{2,4}(9) &= \frac{1}{12}\zeta(2) + \frac{3}{16}. \end{aligned} \quad (4.26)$$

The centered product expansion begins

$$T_2(n)T_4(n) = \frac{1}{3x_n^4} - \frac{7}{36x_n^6} + \frac{61}{360x_n^8} + O(x_n^{-10}). \quad (4.27)$$

Theorem 4.3 consequently gives the following ordinary convergent identities:

$$\sum_{n \geq 0} T_2(n)T_4(n) = \frac{15}{2}\zeta(5) - 3\zeta(2)\zeta(3), \quad (4.28)$$

$$\sum_{n \geq 0} x_n^2 T_2(n)T_4(n) = \frac{1}{4}\zeta(2)\zeta(3) + \frac{1}{6}\zeta(2) + \frac{1}{2}\zeta(3) - \frac{5}{8}\zeta(5), \quad (4.29)$$

$$\sum_{n \geq 0} \left[x_n^4 T_2(n)T_4(n) - \frac{1}{3} \right] = -\frac{7}{80}\zeta(2)\zeta(3) - \frac{1}{30}\zeta(2) - \frac{1}{4}\zeta(3) + \frac{7}{32}\zeta(5) - \frac{3}{40}. \quad (4.30)$$

One further example is

$$\begin{aligned} &\sum_{n \geq 0} \left[x_n^6 T_2(n)T_4(n) - \frac{x_n^2}{3} + \frac{7}{36} \right] \\ &= \frac{31}{448}\zeta(2)\zeta(3) - \frac{1}{672}\zeta(2) + \frac{7}{32}\zeta(3) - \frac{155}{896}\zeta(5) + \frac{31}{672}. \end{aligned} \quad (4.31)$$

Since $T_2 = \psi'$ and $T_4 = \psi'''/6$, multiplication by 6 turns these into identities for products of two unequal polygamma orders. They equally evaluate the centered harmonic numerator $(H_n^{(2)} - \zeta(2))(H_n^{(4)} - \zeta(4))$.

4.5 An analytic generator for all moment orders

For a fixed ordered pair (a, b) , define

$$G_b(t) = \frac{1}{2} \coth(t/2) \sum_{j=1}^{b/2} \zeta(b-2j) \frac{t^{2j}}{(2j)!}. \quad (4.32)$$

Its apparent singularity at zero is removable. It is odd and analytic on $|t| < 2\pi$.

Theorem 4.5 (All-moment generating function). *Let $A_{a,b}$ be the unique analytic solution on $|t| < 2\pi$ of*

$$A_{a,b}^{(a)}(t) = G_b(t), \quad (4.33)$$

whose Taylor polynomial of degree less than a is

$$\sum_{\substack{1 \leq d < a \\ d \text{ odd}}} \left[\zeta(b, a-d) + \frac{1}{2}\zeta(a+b-d) \right] \frac{t^d}{d!}. \quad (4.34)$$

Then the exponential series

$$\mathcal{V}_{a,b}(t) = \sum_{M=0}^{\infty} V_{a,b}(M) \frac{t^{2M}}{(2M)!} \quad (4.35)$$

converges for $|t| < 2\pi$ and satisfies

$$\boxed{\mathcal{V}_{a,b}(t) = \frac{A_{a,b}(t) + A_{b,a}(t)}{2 \sinh(t/2)}}. \quad (4.36)$$

The quotient has a removable value at zero and is even analytic.

Proof. Anchored integration of the analytic function G_b gives existence and uniqueness. Its oddness, and the even value of a , show that $A_{a,b}$ is odd. The Bernoulli generating identity

$$C(t) = \frac{t}{2} \coth(t/2) = \sum_{k \geq 0} \frac{B_{2k}}{(2k)!} t^{2k}$$

gives

$$G_b(t) = C(t) \sum_{j=1}^{b/2} \zeta(b-2j) \frac{t^{2j-1}}{(2j)!}.$$

For positive odd p , its coefficient of $t^p/p!$ is

$$\sum_{j=1}^{\min(b/2, (p+1)/2)} \frac{\binom{p+1}{2j} B_{p+1-2j}}{p+1} \zeta(b-2j).$$

Together with the initial data, this says exactly that

$$A_{a,b}(t) = \sum_{\substack{d \geq 1 \\ d \text{ odd}}} K_{a,b}(d) \frac{t^d}{d!}. \quad (4.37)$$

Finally,

$$\frac{1}{2 \sinh(t/2)} = \frac{1}{t} \sum_{k \geq 0} \frac{B_{2k}(1/2)}{(2k)!} t^{2k}.$$

Multiplication with the sum of the two series in (4.37) gives precisely the coefficient formula (4.12). The numerator is odd and vanishes at zero; the denominator has a simple zero there and no other zero in this disk. Hence the quotient is even analytic throughout the disk, proving actual convergence of (4.35) as well as the identity. $\square$

This analytic generator concerns the regularized values already fixed by (4.6). It is not obtained by interchanging an exponential series with the divergent unmodified moment sums.

4.6 Explicit polylogarithmic primitives and their constants

For an integer $k \geq 1$, set

$$L_k(t) = \int_0^t \frac{u^k}{e^u - 1} du. \quad (4.38)$$

The integrand is analytic at zero. Initially in $\Re t > 0$, the standard polylogarithm differentiation rule [10, §25.12] gives the explicit anchored formula

$$L_k(t) = k!\zeta(k+1) - \sum_{h=0}^k \frac{k!}{(k-h)!} t^{k-h} \text{Li}_{h+1}(e^{-t}). \quad (4.39)$$

For real $t > 0$, all polylogarithms have arguments in $(0, 1)$; this fixes the branch. The formula extends to the right half-plane by analyticity. Differentiating the finite sum makes its adjacent terms cancel and leaves $t^k/(e^t - 1)$. As $t \rightarrow 0$ from the right, all positive powers multiplying the polylogarithms vanish, including $t^k \text{Li}_1(e^{-t})$, and the remaining term cancels $k!\zeta(k+1)$. This proves both the derivative and the anchoring constant. The completed expression, taken as a whole, then continues analytically to $|t| < 2\pi$ through its integral definition. Individual polylogarithm terms need not be assigned separate analytic values around the logarithmic point $t = 0$.

For positive even m , define

$$\begin{aligned} R_{a,m}(t) &= \frac{t^{a+m}}{2(a+m)!} \\ &\quad + \frac{1}{m!(a-1)!} \sum_{r=0}^{a-1} (-1)^r \binom{a-1}{r} t^{a-1-r} L_{m+r}(t). \end{aligned} \quad (4.40)$$

Expanding the anchored integration kernel $(t-u)^{a-1}$ shows that

$$R_{a,m}(t) = \frac{1}{(a-1)!} \int_0^t (t-u)^{a-1} \frac{1}{2} \coth(u/2) \frac{u^m}{m!} du. \quad (4.41)$$

Here one uses $\frac{1}{2} \coth(u/2) = \frac{1}{2} + (e^u - 1)^{-1}$; the first term gives the polynomial in (4.40), and the second gives the finite sum of L integrals. Consequently

$$A_{a,b}(t) = (\text{the polynomial in (4.34)}) + \sum_{j=1}^{b/2} \zeta(b-2j) R_{a,2j}(t). \quad (4.42)$$

Equations (4.23)–(4.24) and (4.36)–(4.42) express the whole moment sequence using ordinary zeta constants, elementary functions, and finitely many $\text{Li}_k(e^{-t})$ with $k \leq a+b$. Every repeated primitive is anchored at zero. Near zero, the ODE or Taylor form is more suitable for numerical evaluation because the separate polylogarithmic terms undergo strong cancellation.

4.7 Scope of the result

The theorem resolves the two-factor even-tail sector of the source question. It does not assert ordinary-zeta closure for the first spectral derivative $F'_{a,b}(-2M)$, independent order derivatives, or products of four tails. In particular, adding a spectral derivative introduces a global logarithmic moment; the finite-value theorem alone does not eliminate it. Verification records and precise next questions are given in Sections 5 and 6.

5 Source audit, normalization safeguards, and verification

5.1 Inspected baseline and source attribution

The source anchor is the ProveIt revision `7bd45777a8a127e5dc69aff8887ca36a79a3059d`, including the eleven archives then present in `docs/incoming` [1]. The archive inventory, immutable source URLs, and hashes accompany the delivery. The mathematical review concentrated on the relative Hurwitz definitions, nonpositive-index reductions, one-free-order generator, centered moment formulas, and the explicit R1, R7, and R8 questions of the incoming harmonic-parity report [2, 3]. This is a targeted analytic review with a complete archive inventory; it is not a line-by-line certification of every proof in the entire repository.

No newly confirmed false theorem in the inspected source material emerged from this review. The refinements below specify hypotheses and normalizations that must survive integration. In particular, the source’s warning against differentiating a fixed deleted index was correct: Section 3 supplies the extra moving-parameter identity needed to cross that boundary.

The kernel attribution deserves explicit preservation. The geometric kernel reorganized in (2.1), and the original entire interpolation that it represents, are established in Ihara, Nakamura, and Yamamoto [5, Proposition 3.3 and Theorem 4.1]. The published reference is *The Ramanujan Journal* **67** (2025), article 1, doi:10.1007/s11139-025-01045-2; the arXiv version remains a useful accessible precursor. Bibliographic entries that describe this work only as a preprint can therefore be updated. The incoming independent-orders report already identifies the interpolant [3]. Our contribution is the stated joint twist completion, the all-block unit-polydisc theorem, and the transverse formulas, rather than a new discovery of their underlying geometric interpolation.

Likewise, the finite gap polynomials and the one-free-order generating function retain their attribution to [2]. Euler’s odd-weight double-zeta reduction remains a classical ingredient, with the explicit primary proof cited in [6]. Several incoming reports have overlapping parity and quartic-zeta results from earlier common baselines. Their separate proofs may be useful, but the canonical question ledger should count the corresponding resolved question once and retain each report’s actual scope.

5.2 An absolutely convergent boundary series can have a different value

The initial convergence domain in Theorem 3.1 is essential. An isolated exponent can cancel the leading asymptotic term of a paired summand, producing an ordinary convergent series whose sum differs from the continuation of the original family. Here is a complete elementary example, requiring no numerical regularization.

Take $m = r = 0$, $a = 2$, and $b = 1$. Then

$$\ell_{0,0}(x) = \zeta(0, x) = \frac{1}{2} - x, \quad h_{2,1}(s) = 1.$$

On the initial half-plane $\Re s > 2$, the defining weighted difference is

$$D_{0,0}(s; 2, 1) = \sum_{n \geq 0} \{f_s(n+1) - f_s(n+2)\}, \quad f_s(x) = \frac{1}{2}x^{-s} - x^{1-s}. \quad (5.1)$$

Its N th partial sum is exactly

$$-\frac{1}{2} - \frac{1}{2}(N+1)^{-s} + (N+1)^{1-s}. \quad (5.2)$$

Consequently its entire continuation is the constant

$$D_{0,0}(s; 2, 1) = -\frac{1}{2}. \quad (5.3)$$

But substituting $s = 1$ in the individual paired summands first gives the ordinary absolutely convergent series

$$\sum_{n \geq 0} \frac{1}{2} \left(\frac{1}{n+1} - \frac{1}{n+2} \right) = \frac{1}{2}. \quad (5.4)$$

The values differ by one, which is the boundary term $(N+1)^{1-s}$ in (5.2). This discrepancy does not contradict analytic continuation: the paired series lacks locally normal convergence on a complex neighborhood of $s = 1$.

Thus (3.7) and (3.8) must retain the phrase “the continuation from the stated initial domain.” Literal summation at an accidentally improved boundary point is a different operation. The example is also a useful negative control for software that evaluates parameterized infinite sums after specialization.

5.3 Hypotheses and constants to preserve during integration

Moving and fixed indices. The polynomial identity for a word with a letter fixed at $u = -m$ contains no information about derivatives in the transverse variable u . It can be differentiated in its independently free letters. To differentiate in u , first use the entire identity (3.12), and then specialize. The functions $\ell_{m,r}$ are the additional data introduced by that operation. The finite-family bound in Proposition 3.7 concerns one marked index; it does not establish the same bound for several independently marked indices.

Strict endpoints and Bernoulli signs. For a strict unit-lattice gap the finite power sum is

$$\sum_{z < x < y} x^m = \frac{B_{m+1}(y) - B_{m+1}(z+1)}{m+1}.$$

Accordingly $B_1 = -1/2$ and $B_1(1) = 1/2$ occur in different positions. At $m = 0$, the missing endpoints account for the last term in

$$\mathcal{H}(s, 0, t) = \mathcal{H}(s-1, t) - \mathcal{H}(s, t-1) - \mathcal{H}(s, t).$$

The analogous term in (3.18) must also remain after transverse differentiation. Its sign is fixed by the Hurwitz shift identity, not by a choice of regularization.

Centered cutoff constants and the two zero-order zeta values. The mixed-tail proof uses the cutoff $0 \leq n < N$ and the constant term in the variable N . The equality with the spectral value is proved in (4.9); it is not a general rule for every cutoff coordinate. Two different zero-order zeta values appear:

$$\zeta(0, 1/2) = 0, \quad \zeta(0) = -\frac{1}{2}.$$

The first makes the continued sums of the subtracted centered even polynomials vanish. The second encodes the surviving rational boundary term in (4.22). Dropping either distinction changes the moment formula. At $d = a + b - 1$, the two oriented $\gamma/2$ terms cancel the diagonal $-\gamma$; an isolated $\zeta(1)$ is never introduced.

Domains, branches, and spanning statements. Endpoint parameters stay in the right half-plane. The twists satisfy the consecutive-sum inequalities in (2.2); removable zero collisions do not remove the remaining nonzero-period hyperplanes. The polylogarithmic primitives are first fixed in the right half-plane and then continued as their complete anchored expressions, as in (4.39). The finite list in (4.25) is a proved rational spanning list. Its at-most-max(a, b) count is not an assertion of arithmetic independence or minimality. The ordinary moment theorem also does not imply ordinary-zeta closure of its first spectral derivative, which contains additional global logarithmic data.

5.4 Reproducible verification and the limits of its evidence

The analytic proofs above establish the unrestricted identities. The companion scripts provide finite exact checks and independent high-precision diagnostics of the most sensitive formulas. The recorded coverage is summarized below.

Family	Exact	Numerical	What is compared
Geometric kernels and zero blocks	590	0	Endpoint composition, deconcatenation, kernel recurrence, finite intervals, inverse intervals, gap coefficients, and insertion sums.
Mixed-tail moments	460	30	400 boundary identities and 60 finite summation-by-parts identities; direct Hurwitz heads and centered tail expansions at two cutoffs.
Transverse identities	0	15	Finite Gamma/Hurwitz gaps, noninteger endpoint tails, Barnes values, negative-outer-order formulas, and the boundary discrepancy above.
Normalized Mellin derivatives	0	24	Subtracted logarithmic quadrature versus derivatives of explicit elementary Mellin transforms for two independent test functions.
Moment generators	0	3	Anchored quadrature, explicit polylogarithmic primitives, and a Taylor sum using 65 exact moment coefficients in each comparison.
Total	1,050	72	Finite algebra and numerical diagnostics have different evidentiary roles.

The 590 kernel checks use rational arithmetic only. Positive fourth-power exponential nodes and quarter-integral endpoints permit exact tests even when $a - b$ is nonintegral. The 460 moment checks are finite identities over rational symbolic coefficient rings. In particular, the finite summation-by-parts tests keep the two tail constants symbolic.

The transverse diagnostics use 65 decimal working digits and include the discrepancy (5.3)–(5.4) as a deliberate boundary control. A separate 80-digit test of (3.29) uses two quadratic-polynomial exponential functions, one with nonreal coefficients and decay parameter. For each function, the centers $m = 0, 1, 3$ and derivative orders $r = 1, 2, 3, 4$ compare the subtracted logarithmic integral with a numerical derivative of its known elementary Mellin transform. All 24 cases satisfy the scripted 10^{-50} tolerance; the largest observed discrepancy is about 3.10×10^{-79} . These functions test the normalization lemma independently of the relative harmonic kernel.

The moment diagnostics use 95 working digits, with orders through (6, 8) and moment indices through $M = 7$. They evaluate a direct finite Hurwitz-zeta head and a separately assembled centered asymptotic tail at $(N, K) = (48, 32)$ and $(64, 36)$. The largest observed discrepancy at the larger cutoff is about 1.16×10^{-63} . The three generator comparisons include a nonreal parameter; their largest recorded Taylor-sum discrepancy is about 1.08×10^{-89} . The 65 Taylor coefficients are exact inputs to those three numerical comparisons, not 65 additional independent exact tests.

These residuals are observed floating-point discrepancies, not rigorous error bounds or interval enclosures. Working precision alone is not an accuracy guarantee. The recorded diagnostics cover the stated examples; they do not constitute numerical quadrature of every multidimensional integral in Theorem 3.5. Uniform integrability, simultaneous branch cancellation, and continuation at arbitrary orders are established by the proofs, not by a finite test count. The delivery includes neither a proof-assistant formalization nor an external referee report.

5.5 Status of the broader conjectures

The contribution ledger records R7 as answered in the specified arbitrary-depth zero-decoration family, R8 as answered for one marked index and all of its nonpositive-center derivatives, and R1 as answered for the full two-factor even-tail sector. Extra harmonic factors, several independently marked indices, and additional global spectral derivative reductions remain separate questions.

The Gaussian S_6 and revised S_8 candidates retain their conjectural status [1, 2]. No new functional identity or exact period certificate for either candidate is supplied here. Existing nonmembership results concern their explicitly named formal relation spaces and must not be read as period nonvanishing theorems. The next research questions in Section 6 begin at these stated boundaries.

6 Further exact research questions

The results above change the starting point of the next investigation. The all-zero-block generator and the single marked transverse derivative need no further conjectural existence statement. The unresolved issues concern additional factors, several marked positions, arithmetic reduction, and analytic continuation beyond the domains proved here. The questions below specify an object to compute and a criterion for progress. They are proposals, not assertions that the suggested reductions hold.

6.1 Higher products and logarithmic moments

Q1. Four and more even-order tails. For an even number $2d \geq 4$ of even orders $a_j \geq 2$, consider

$$F_a(s) = \sum_{n \geq 0} (n + 1/2)^{-s} \prod_{j=1}^{2d} \zeta(a_j, n + 1).$$

The centered expansion has exponents $2d - \sum_j a_j - 2k$, so the same subtraction argument makes all even spectral integers regular. The open arithmetic problem is to evaluate every $F_a(-2M)$ in an explicit finite list of multiple-zeta coordinates whose size is independent of M . The first useful case is $(a_1, a_2, a_3, a_4) = (2, 2, 2, 4)$. Finite summation by parts and ordering the surviving indices give a concrete route, but the two-factor Euler reduction does not by itself remove every higher-depth coordinate. A satisfactory theorem should give the finite coordinate map first and prove every subsequent reduction. For an odd number of even tails, possible poles occur at even spectral integers; that problem additionally requires a specified Laurent finite part and is a different extension.

Q2. Which logarithmic remainders cancel? Corollary 4.2 gives a convergent exact formula for every $F_{a,b}^{(r)}(-2M)$, including its finite Hurwitz correction. The remaining question is whether there are nontrivial, explicitly specified rational linear combinations of these derivatives whose logarithmic remainder sums cancel or reduce to endpoint data. Begin with $r = 1$, $(a, b) = (2, 4)$ and $(2, 2)$, and

retain the $-\log 2/6$ term in (4.8). One should seek an identity at the level of the summands or a parameter-dependent kernel; an integer relation among a few decimal values would only suggest a target. Independent differentiation of a tail order, such as $\partial_a F_{a,b}$ before specializing a , is another problem because it differentiates both the Hurwitz tail and its asymptotic coefficients.

Q3. A smallest proved span for each pair of orders. Corollary 4.4 supplies at most $\max(a, b)$ coordinates for the entire sequence. Determine all reductions of that list obtainable from stated algebraic identities, and give an algorithm producing a smaller *proved* spanning list. There are two distinct questions: the rank of explicit rational coefficient vectors in a chosen formal zeta-product alphabet, and arithmetic independence of the resulting numbers. Only the first is a finite linear-algebra computation without additional period theorems. A uniform description for $(2, 2q)$ would be a useful initial result.

6.2 Several moving positions and primitive calculus

Q4. Two independently marked positions. Construct a jointly continued gap formula for

$$\partial_u^r \partial_v^q \mathcal{H}_{a,b}(s, u, v, t) \big|_{u=-m, v=-n}, \quad m, n \geq 0, \quad r, q \geq 1,$$

and identify a finite family of decorated lower-depth objects that remains sufficient after arbitrary unmarked nonpositive deletions. The adjacent case $m = n = 0, r = q = 1$ has a precise finite-gap guide. For two surviving positive real lattice values $y > z$ in a finite unit-lattice interval, put

$$P_r(y, z) = \ell_{0,r}(z+1) - \ell_{0,r}(y).$$

The elementary symmetric-polynomial identity gives

$$\sum_{y>i>j>z} \log i \log j = \frac{1}{2}(P_1(y, z)^2 - P_2(y, z)).$$

Here P_1 is the negative sum of the gap logarithms and P_2 is their squared-logarithm sum. This finite identity suggests the shape of a Gamma–Stieltjes formula. A theorem at arbitrary endpoints must still identify its decorated continuation with the prescribed relative quotient. The separate-position and unequal-center cases may require more than products of single-gap primitives.

Q5. Ordered versus symmetrized transverse data. Equation (3.26) evaluates the sum of the three possible insertion positions using depth-two data. Determine an exact decomposition of several marked derivatives into their symmetrized part and the remaining ordered part. At two marked positions, stuffle products provide explicit relations but do not identify all ordered differences. A useful result would describe a finite basis of these ordered differences modulo the stated stuffle and deletion relations, with the endpoint normalization retained throughout.

Q6. Endpoint differentiation and normalized antiderivatives. Combine the transverse calculus with differentiation in a and b . The identities

$$\partial_x \zeta(u, x) = -u \zeta(u+1, x), \quad \mathcal{B}'_m(x) = \mathcal{B}_{m-1}(x)$$

suggest a closed transport system connecting moving harmonic orders, polygamma factors, and the balanced primitive ladder. Determine that system for a depth-three word with one marked position, then integrate it between specified endpoint values. The target is a formula whose Gamma or Barnes constants are fixed by the same relative quotient, rather than by separately chosen indefinite integrals. At $r \geq 2$, the analogous system involves generalized Stieltjes primitives and should state their additive normalization explicitly.

6.3 Rational shifts and continuation of the generators

Q7. Common rational shifts and colored tails. For rational $c > 1/2$, replace the moment family by

$$\sum_{n \geq 0} (n + c - 1/2)^{-s} \zeta(a, n + c) \zeta(b, n + c).$$

Its centered parity remains available, but the removed polynomial terms now have continued values $\zeta(-2j, c - 1/2)$, which generally do not vanish. Establish the exact analogue of the finite Bernoulli formula with all these counterterms. Then determine the appropriate colored or cyclotomic multiple-zeta reduction for rational c . Unequal shifts in the two tails should be treated separately, since they no longer share the same centered expansion. Any differentiation of rational multiplication formulas must keep the conductor powers and their logarithmic derivatives.

Q8. Rational endpoint Gamma–Stieltjes identities. Specialize the finite endpoint formula (3.31) to $a, b \in \mathbb{Q}_{>0}$ and apply Hurwitz distribution and character projections. For example,

$$\sum_{j=1}^q \zeta(s, j/q) = q^s \zeta(s)$$

gives a completely specified hierarchy after differentiation at $s = -m$. Determine which linear combinations of transverse weighted tails reduce to derivatives of Dirichlet L -functions and logarithms of Gamma or Barnes values, and which still require ordered coordinates. The equality of the finite endpoint expressions provides an exact starting point; no conjectural independence of Stieltjes constants is needed for this transport step.

Q9. Singular hyperplanes beyond the common twist tube. Theorem 2.3 proves joint holomorphy on Ω_k . The kernel identifies the only remaining candidate singular hyperplanes as nonzero exponential periods of consecutive sums. Determine their residues or local branching, first at depth two, including further cancellations at specific nonzero periods. For integral $a - b$, the function z^{a-b} is single valued near every nonzero node, so all finite node collisions are removable; for $a - b = N \geq 0$ the finite exponential sum gives the same conclusion. For nonintegral endpoint differences, a precise local continuation theorem would explain which parts of the unit decoration polydisc can be enlarged and on which sheets. It should treat the complete relative expression, since separately continued Lerch summands can conceal cancellations.

Q10. Functional identities for the all-moment generator. The ODE and polylogarithmic representation in Theorem 4.5 reduce each fixed pair of orders to finitely many anchored primitives. Seek recurrences linking different pairs (a, b) , reflection or continuation identities in t , and exact special values at nonzero arguments where the polylogarithms have a useful algebraic argument. The common radius 2π also makes the nearest singularities at $\pm 2\pi i$ accessible. Their local expansions may give exact coefficient relations or full asymptotic expansions of the moment sequence. Estimates alone are a secondary goal here; the primary target is a new identity between the generators.

6.4 Integration with the existing conjectural program

Q11. An analytic bridge to the Gaussian S_6 and S_8 candidates. The source manuscript distinguishes its formal Cayley or period coordinates from proved evaluations. The new generators should be compared with the existing Gaussian kernels only through an explicit analytic transformation: a convergent integral identity, a justified parameter deformation, or a proved functional equation. Determine whether such a transformation sends a specific conjectural remainder to a Gamma–Stieltjes gap or a mixed-tail coordinate evaluated here. Until that bridge is proved, the present exact results supply neither a proof nor a refutation of the S_6 or revised S_8 formulas.

Q12. Formalization of the exact algebra before the analytic layer. The divided-difference product identity, finite endpoint composition, strict-gap binomial coefficients, Bernoulli deletion, and finite summation-by-parts formulas are suitable first formalization targets. Their statements should use exact rational or symbolic arithmetic and retain the boundary terms. The analytic layer then needs explicitly stated normalized-Mellin continuation, local normal convergence, and Hurwitz asymptotics. The accompanying programs are reproducible checks of finite instances; they do not replace those quantified theorems. A useful integration milestone is a mechanically checked finite algebraic core with the analytic lemmas listed as precise dependencies.

For an efficient next investigation, the most concrete starting points are the four-tail example in Q1, the adjacent two-marked gap in Q4, and the rational-shift extension in Q7. Each starts from a finite identity or a convergent kernel already available here. They also test three different potential obstructions: higher arithmetic depth, nonlinear ordered primitives, and nonvanishing endpoint counterterms.

References

- [1] ProveIt Contributors, *Polylogarithms and their Arithmetic Bridges*. ProveIt repository, pinned manuscript and editorial-ledger snapshot, 11 October 2026. <https://github.com/VladimirReshetnikov/ProveIt/tree/7bd45777a8a127e5dc69aff8887ca36a79a3059d/Analysis/Polylogarithms/docs/manuscript>.
- [2] ProveIt research collection, *Harmonic Parity and Resolvent Identities: Exact Lerch generators, Stieltjes primitives, and the quartic and higher Tornheim layers*. Research report, 11 October 2026; in particular, the centered harmonic calculus, gap-deletion formulas, and research questions R1, R7, and R8. Incoming archive at the revision of [1]: https://github.com/VladimirReshetnikov/ProveIt/blob/7bd45777a8a127e5dc69aff8887ca36a79a3059d/docs/incoming/ProveIt_Harmonic_Parity_and_Resolvent_Identities_2026-10-11.zip.
- [3] ProveIt research collection, *Independent Orders and Exact Reductions in Polylogarithm–Zeta Calculus*. Research report, 11 October 2026; in particular, the entire relative Hurwitz interpolant and its nonpositive-index reductions. Incoming archive at the revision of [1]: https://github.com/VladimirReshetnikov/ProveIt/blob/7bd45777a8a127e5dc69aff8887ca36a79a3059d/docs/incoming/ProveIt_Independent_Orders_2026-10-11.zip.
- [4] ProveIt research collection, *Higher Reflection Identities and Resonant Zeta Calculus*. Research report, 11 October 2026; in particular, the centered harmonic parity and resonant-continuation conventions. Incoming archive at the revision of [1]: https://github.com/VladimirReshetnikov/ProveIt/blob/7bd45777a8a127e5dc69aff8887ca36a79a3059d/docs/incoming/ProveIt_Higher_Reflection_Identities_2026-10-11.zip.

- [5] Kentaro Ihara, Yayoi Nakamura, and Shuji Yamamoto, *Interpolant of truncated multiple zeta functions*. The Ramanujan Journal **67** (2025), article 1. <https://doi.org/10.1007/s11139-025-01045-2>. Author preprint: <https://arxiv.org/abs/2407.20509>. Especially Proposition 2.1, Proposition 3.3, Theorem 4.1, and Corollary 4.2.
- [6] David M. Bradley, *A signed analog of Euler’s reduction formula for the double zeta function*. Author preprint, arXiv:0707.4486 (2007). <https://arxiv.org/abs/0707.4486>. The unsigned specialization is the classical odd-weight Euler reduction; see equations (2)–(3) and Proposition 1.
- [7] National Institute of Standards and Technology, *NIST Digital Library of Mathematical Functions*, §25.11, Hurwitz zeta function. In particular, the shift law, polygamma and Bernoulli special values, parameter differentiation, and Lerch’s log-Gamma identity in equations 25.11.3, 25.11.12, 25.11.14, 25.11.17, and 25.11.18; the asymptotic expansion is 25.11.43. Accessed 11 October 2026. <https://dlmf.nist.gov/25.11>.
- [8] National Institute of Standards and Technology, *NIST Digital Library of Mathematical Functions*, Chapter 5, Gamma function; especially §5.2 (definitions), §5.7 (series expansions), §5.11 (asymptotic expansions), and §5.15 (polygamma functions). Accessed 11 October 2026. <https://dlmf.nist.gov/5.2>; <https://dlmf.nist.gov/5.7>; <https://dlmf.nist.gov/5.11>; <https://dlmf.nist.gov/5.15>.
- [9] National Institute of Standards and Technology, *NIST Digital Library of Mathematical Functions*, §24.4, Basic properties of Bernoulli and Euler polynomials. In particular, the difference, reflection, power-sum, and finite-expansion identities in equations 24.4.1, 24.4.3, 24.4.9, and 24.4.12. Accessed 11 October 2026. <https://dlmf.nist.gov/24.4>.
- [10] National Institute of Standards and Technology, *NIST Digital Library of Mathematical Functions*, §25.12, Polylogarithms. Accessed 11 October 2026. <https://dlmf.nist.gov/25.12>.
- [11] National Institute of Standards and Technology, *NIST Digital Library of Mathematical Functions*, §25.14, Lerch’s transcendent. Accessed 11 October 2026. <https://dlmf.nist.gov/25.14>.
- [12] National Institute of Standards and Technology, *NIST Digital Library of Mathematical Functions*, §5.17, Barnes’ G -function (double gamma function). In particular, the recurrence, normalization, product, and logarithmic integral formulas in equations 5.17.1, 5.17.3, and 5.17.4. Accessed 11 October 2026. <https://dlmf.nist.gov/5.17>.
- [13] Olivier Espinosa and Victor H. Moll, *A generalized polygamma function*. Integral Transforms and Special Functions **15** (2004), 101–115. Author preprint: <https://arxiv.org/abs/math/0305079>. The balanced negapolygamma normalization supplies the fixed constants used in the primitive ladder.
- [14] Ilan Vardi, *Determinants of Laplacians and multiple gamma functions*. SIAM Journal on Mathematical Analysis **19** (1988), 493–507. <https://doi.org/10.1137/0519035>. The Hurwitz-zeta derivative identity fixes the Barnes- G normalization used in the endpoint formula.